PROBLEM STATEMENT :-

FESTOS

Aarohan is a three-day inter-college festival run through disconnected spreadsheets, forms, calls and messaging groups. The core team needs one internal control portal that can coordinate participants, accommodation, events, venues, logistics, merchandise, food vendors and staff access.

Your 24-hour challenge is to build FestOS: a working portal that makes defensible decisions under conflicting constraints and remains useful when the plan changes. The data is deliberately imperfect. State assumptions, preserve uncertainty and make every important override visible.

EPINOIA 2026 | 24-HOUR HACKATHON | IIM INDORE



The Situation

Build the operating system for a festival that cannot pause.

Aarohan expects roughly 2,400 participants from 38 colleges, 31 events across nine venues, 14 food vendors and a core team working in shifts. Participants arrive over five days and may compete in several events.

THE CURRENT FAILURE

Information lives in separate files and conversations. A delayed bus, room change or equipment failure can affect check-in, accommodation, transport, food, merchandise and event attendance within minutes. No single person can see the full consequence of a change.

YOUR MANDATE

Create one internal portal for the people running the festival. It is not a participant registration website. It should help organizers verify facts, plan constrained resources, act during disruptions and explain why a decision was made.

2,400	31	9
EXPECTED PARTICIPANTS	EVENTS	VENUES

What the Fest Looks Like

Every module affects another module.

PARTICIPANTS AND CONTINGENTS

Contingents range from 6 to 140 members. Participants may enter several solo or team events, arrive at different times and carry accessibility, dietary, accommodation and merchandise requirements.

ACCOMMODATION

Demand exceeds low-cost supply. On-campus rooms, external properties, transport windows, check-in, check-out and cleaning readiness must be considered together.

EVENTS AND VENUES

Rounds, participant clashes, venue capacity, judges, volunteers, equipment, setup, teardown and travel time all constrain the schedule.

LOGISTICS AND MERCHANDISE

Stock is limited, moves between locations and may be unavailable because it is reserved, in transit, damaged or awaiting return. Merchandise must be issued once to the right person.

FOOD VENDORS

Menus, dietary coverage, hygiene certificates, operating hours, capacity, rates and payment terms differ. Vendor interruptions can create immediate service and safety gaps.

A change is complete only when its downstream effects are visible.

Accommodation Rules

Hard needs must not disappear inside an optimization score.

1. Respect declared accessibility and privacy needs. Accessibility requirements are hard constraints.
2. Support gender-segregated housing where requested without forcing people outside the binary into an unsuitable category.
3. Keep contingents together where possible. If a split is necessary, minimize it and make the reason explicit.
4. Place each contingent leader near the group. House faculty or staff escorts separately but nearby.
5. Respect room capacity, readiness, cleaning, check-in and check-out windows.
6. Account for early or late events, external travel time, transport availability and safety coverage.
7. Show the cost of every external placement and stay within budget unless an authorized user overrides it.
8. Never interpret a blank field as consent, eligibility, gender, dietary clearance or absence of need.

REQUIRED BEHAVIOR

The system may recommend an allocation, but it must show violated or relaxed constraints, explain trade-offs and require a reason for high-risk overrides. There is no single perfect allocation; there are only defensible ones.

What You Must Build

A portal that only stores rows is a spreadsheet with a login page.

- A unified participant record connecting contingent, events, accommodation, merchandise, food needs and operational status.
- Accommodation allocation that respects hard rules, exposes shortages and explains cost-versus-cohesion trade-offs.
- Event and venue scheduling that detects participant, team, judge, volunteer, venue and equipment conflicts.
- Time- and location-aware tracking for logistics and merchandise inventory.
- A usable vendor view covering dietary capacity, compliance, interruptions and settlement.
- Role-based access, audit history and protected handling of contact, medical, accessibility and room data.
- A control view that surfaces what needs attention before someone has to call the festival director.

DESIGN FREEDOM

Use any stack or interface. Optimization, rules, forecasting, agents and language models are allowed. Technology will be judged only through the quality, clarity and resilience of the decisions it enables.

The Data Pack

Operational data is messy by design. Read it critically.

The narrative expects about 2,400 participants. The pack contains 2,836 raw participant records. Do not hide the mismatch. Reconcile, flag or explain it.

participants.csv
2,836 records

registrations.csv
5,580 mappings

contingents.csv
38 contingents

events.csv
31 events

venues.csv
9 venues

core_team.csv
48 members

hostels.csv
200 rooms / 710 beds

external_stays.csv
6 properties / 740 beds

inventory_logistics.csv
72 lines

inventory_merch.csv
10 lines

vendors.csv
14 vendors

README.txt
data guidance

HOW TO USE IT

Expect incomplete fields, inconsistent labels, timing anomalies, duplicate-looking records, capacity pressure and cross-file conflicts. State assumptions, preserve uncertainty, validate imports and allow manual overrides with reasons.

Rules and Deliverables

Build quickly, but remain accountable for what the system does.

RULES OF THE GAME

- AI tools are permitted and encouraged. Your team remains responsible for testing, privacy, security and failure handling.
- Use any language, framework or deployment approach. A locally running product is acceptable.
- Use only synthetic participant data. State assumptions and disclose external dependencies.
- Include one slide stating what AI did, what your team decided and how outputs were checked.

DELIVERABLES

01 **A working portal that judges can interact with.**

02 **A sample accommodation, scheduling and inventory plan generated from the pack.**

03 **A seven-minute pitch followed by three minutes of questions.**

04 **Setup instructions, test credentials and the required AI disclosure.**

LIVE TEST

Judges will inject a disruption such as a late contingent, duplicate record, hostel closure, equipment failure, venue change, vendor shutdown or duplicate merchandise claim. Your response must change real system state and show the consequences.

How You Will Be Judged

Depth and judgment matter more than the number of screens.

20% **OPERATIONAL UNDERSTANDING**
How well the product reflects how a real fest runs and fails.

20% **ALLOCATION AND SCHEDULING LOGIC**
Constraints respected, trade-offs explained and exceptions contained.

20% **ORIGINALITY**
Important problems found and solved beyond the explicit brief.

15% **WORKING DEMO AND ROBUSTNESS**
Reliable flows and a sensible response to the live test.

10% **USABILITY UNDER PRESSURE**
Fast, clear workflows for organizers and volunteers.

10% **DATA ACCESS AND PRIVACY**
Validation, permissions, audit history and sensitive-data handling.

5% **PITCH CLARITY**
Clear choices, evidence, limitations and answers.

HOUR 0	HOUR 12	HOUR 22	HOUR 24
Kickoff	Twist	Code freeze	Presentations